FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

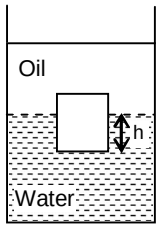
PART – I

SECTION – A

1. CD

Sol. $\left[\rho_w \times 10^{-2} \times h\right] + \left[\rho_o \times 10^{-2}(0.1-h)\right] = 0.92$

$\Rightarrow$ h = 8 cm



2. CD

Sol. Particle *P* will have a velocity in vertical direction.

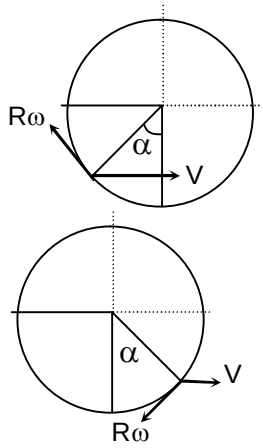
i.e. $R\,\omega\cos\alpha = V$

$$\alpha = \cos^{-1}\frac{V}{R\omega}$$

Required angle θ is $\theta = \pi - \alpha = \pi - \cos^{-1}\frac{V}{R\omega}$

And in second case $R\omega\cos\alpha = V$

$\alpha = \cos^{-1}\frac{V}{R\omega}$, $\theta = \pi + \cos^{-1}\frac{V}{R\omega}$



3. AB

Sol. $-\frac{dv}{dt} = \frac{v^2}{R}$ & $-v\frac{dv}{ds} = \frac{v^2}{R}$

4. ABD

Sol. By conservation of angular momentum,

$I_1\omega_i = (I_1 + I_2)\omega_f$

$\omega_f = \omega_1 / 3$

$E_i = \frac{1}{2} I_1 \omega_i^2$

$E_f = \frac{1}{2}(I_1 + I_2)\omega_f^2$

Ratio of the heat produced to initial kinetic energy $= \frac{E_i - E_f}{E_i} = \frac{2}{3}$

5. BD

Sol. $E \propto m$ and $r \propto \frac{1}{m}$

6. ACD

Sol. $Fx = \frac{1}{2}kx^2$

$\Rightarrow \quad x = \frac{2F}{k}$, $A = \frac{2F}{k} - \frac{F}{k} = \frac{F}{k}$

7. BCD

Sol. dQ = dU + dW ; dQ = 0 in adiabatic process.

8. ABD

Sol. Bernoulli's Th^m for an orifice at depth 'x' in liquid '3d'.

$P_o + \left(dg\frac{H}{2} + 3dg.x\right) = P_o + \frac{1}{2} \times 3d \times v^z$ …(1)

$\frac{H}{2} - x = \frac{1}{2}gt^2$ …(2)

$R = Vt$ …(3)

Solve for R and apply maxima/minima.

9. A

Sol. Degree of function = 3

W. O. = Area PV graph = $\frac{RT_o}{3}$

Charge in internal Energy

$\Delta U = nC_v\Delta T$

$= \frac{f}{2}nR\Delta T = \frac{f}{2}(P_f v_f - P_i V_i)$

$= \frac{3}{2}\left(\frac{3P_o}{2} 2V_o - P_o V_o\right) = 3P_o V_o$

$\Delta U = RT_o$

Heat absorbed $(1 \to 2 \to 3)$

$$\Delta Q = \Delta U + W$$

$$\Delta Q = RT_o + \frac{RT_o}{3}$$

$$\Delta Q = \frac{4T_oR}{3}$$

Heat observer $(1 \to 2)$

$$\Delta Q = \Delta U + W$$

$$= \frac{f}{2}(P_fV_f - P_iV_i) + W$$

$$= \frac{3}{2}(P_0 2V_o - P_oV_o) + P_oV_o$$

$$= \frac{5}{2}P_oV_o = \frac{5}{2}\left(\frac{RT_o}{3}\right)$$

10. D

Sol. Process $1 \to 2$ (Isothermal)

$2 \to 3$ (Isochoric)

W.O. $(1 \to 2 \to 3)$

$$= nRT\ln\left(\frac{V_f}{V_i}\right) + W_{2\to3}$$

$$= \frac{RT_o}{3}\ln 2 + 0$$

ΔU $(1 \to 2 \to 3)$

$$\Delta U = \frac{f}{2}nR(T_f - T_i)$$

$$= \frac{3}{2}R\left(T_o - \frac{T_o}{3}\right)$$

$$\Delta U = RT_o$$

For $1 \to 2 \to 3$

$$\Delta = \Delta U + W$$

$$\Delta Q = RT_o + \frac{RT_o}{3}\ln 2$$

For $1 \to 2$

$$Q = \frac{f}{2}nR(T_f - T_i) + nRT\ln\left(\frac{V_f}{V_i}\right)$$

$$= 0 + \frac{RT_o}{3}\ln 2$$

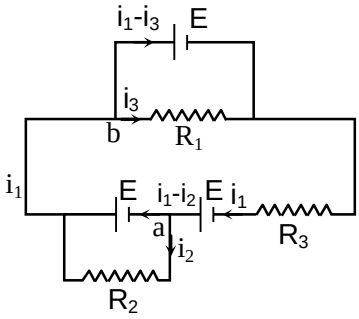
11. C

Sol. On solving by KVL we get, $i_1 = \frac{E}{R}$, $i_2 = -\frac{E}{R}$, $i_3 = \frac{E}{R}$

$\Rightarrow i_1 - i_2 = \frac{2E}{R}$ and $i_1 - i_3 = 0$

So reading of ammeter A_1 is E/R = 4/3

A_2 is 2E/R = 8/3, A_3 is zero and ΔV_{ab} = 4V



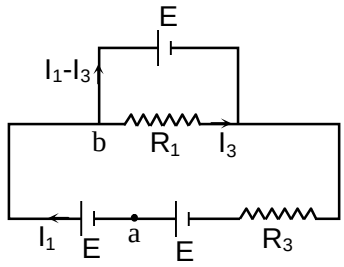
12. A

Sol. Apply KVL

$-I_1 \times 3 + 4 + 4 - 3I_3 = 0$ …(1)

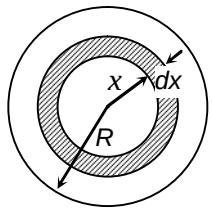
$I_3 = \frac{4}{3}$ …(2)

$I_1 = \frac{4}{3}$



SECTION – C

13. 00002.50



$d\phi = (2\pi x dx)kxt^2$

$\phi = \frac{2}{3}kt^2\pi x^3$

$\frac{d\phi}{dt} = \frac{4k\pi tx^3}{3}$

$E2\pi x = \frac{4\pi ktx^3}{3}$; $E = \frac{2}{3}ktx^2$; $d\tau = \left(\frac{2}{3}ktx^2\right)\frac{Q}{\pi R^2}(2\pi x dx)x$

$\int d\tau = \frac{4}{3}\frac{ktQ}{R^2}\int_0^R x^4 dx$ $\Rightarrow \tau = \frac{4}{3}\frac{ktQ}{R^2}\frac{R^5}{5}$

At t = 15sec, τ = 2.50 Nm

14. 00006.28

Sol. Let elongation of spring be x_0 in equilibrium. Then,

$2T + mg\sin\theta = 2kx_0$ …(i)

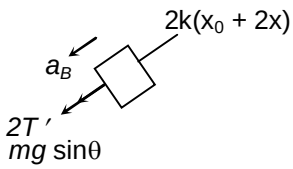
and $T = mg$ …(ii)

Let Block B is displaced by x down the inclination

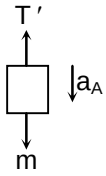
F.B.D. of B

$-ma_B = 2k(x_0 + 2x) - 2T' - mg\sin\theta$ …(iii)

F.B.D. of *A*



$mg - T' = ma_A$



Also, $a_A = 2a_B$

$T' = mg - 2ma_B$

$-ma_B = 2kx_0 + 4kx - 2mg + 4ma_B - mg\sin\theta$

$-ma_B = 4kx + 4ma_B$

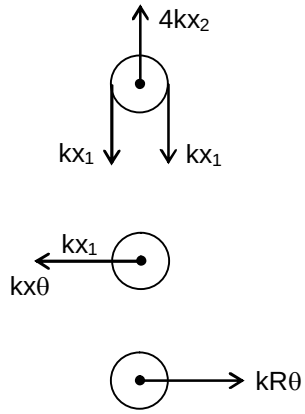
$a_B = -\frac{4k}{5m}x$

$\therefore\ T = 2\pi\sqrt{\frac{5m}{4k}}$

T = 6.28 s.

15. 00008.56

Sol. When the cylinder is displaced towards right, (θ is angle of rotation of diameter of the cylinder about the contact point) then



$x_1 + 2x_2 = R\theta$

$x_1 + x_1 = R\theta$

$x_1 = \frac{R\theta}{2}$

$\tau = -[kx_1R + kR\theta R]$

$\frac{3}{2}mR^2\alpha = -\left[\frac{KR^2\theta}{2} + KR^2\theta\right]$

$\alpha = -\frac{k}{m}\theta \quad ; \quad t_1 = \pi\sqrt{\frac{m}{k}}$

When cylinder moves towards left, then

Torque about the contact point

$-kR^2\theta = \frac{3}{2}mR^2\alpha$

$\alpha = -\left(\frac{2}{3}\frac{k}{m}\right)\theta \quad ; \quad t_2 = \pi\sqrt{\frac{3m}{2k}}$

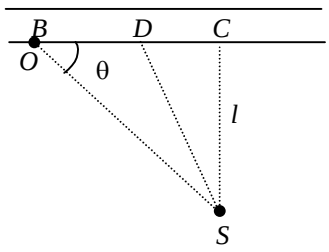
So, time period of oscillation

$T = t_1 + t_2$

$= \frac{\pi}{2}\left(3 + \sqrt{6}\right) = 5.56$ sec.

16. 00136.00

Sol. Let the bus be at O when it sends a signal that is detected by the detector as of frequency = 1500 Hz



$\therefore \qquad f' = \left(\frac{v}{v - v_B\cos\theta}\right)\times 1000 = 1500$

$\therefore \qquad \cos\theta = \frac{\sqrt{3}}{2} \quad \Rightarrow \quad \theta = 30°$

By the time signal reaches at S the bus reaches at D. Let this time be t_0

$\therefore \quad t_0 = \frac{OS}{v} = \frac{l\text{cosec}\,\theta}{v}$ … (1)

Now man fires and the bullet reaches C in time t_1 (say). In the same time bus moves from D to C

$\therefore \quad t_1 = \frac{l}{u}$; where u = speed of bullet

Also, OD + DC = $l\cot\theta$

$v_B t_0 + v_B t_1 = l\cot\theta$

$v_B \left(\frac{l\text{cosec}\,\theta}{v}\right) + v_B (l/u) = l\cot\theta$

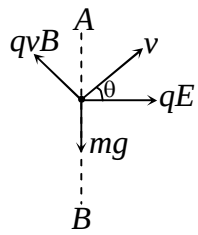
$\frac{2}{3\sqrt{3}} \times 2 + \frac{2}{3\sqrt{3}} \frac{v}{u} = \sqrt{3}$

$\therefore \quad \frac{v}{u} = \frac{5}{2}$

$\therefore \quad u = \frac{2}{5} v = \frac{2}{5} \times 340$ = 00136.00 m/s

17. 00019.60

Sol. When crosses *AB*



$qvB_0 \cos\theta = mg$

$qvB_0 \sin\theta = qE$

$\tan\theta = \frac{qE}{mg} = 1$

$\theta = \frac{\pi}{4}$

along horizontal $v\cos\theta = \frac{qE}{m} t_0$

$u - gt_0 = v\sin\theta$

$u = \left(g + \frac{qE}{m}\right) t_0 = 2gt_0$

18. 00002.00

Sol. $\int \tau dt = NAB \int i dt$

$I\omega_0 = NABQ$ ……(i)

$\frac{1}{2} I\omega_0^2 = \frac{1}{2} C\theta_{max}^2$ …….(ii)

From equation (i) and (ii)

$\theta_{max} = \frac{NABQ}{\sqrt{CI}}$

Chemistry

PART – II

SECTION – A

19. AC

Sol. In BF_3 due to $p\pi$- $p\pi$ back bond and existence of pi-bond in CO_2.

20. A

Sol. In B & C free radical intermediates are formed. In (D) no full-fledged intermediate is formed.

21. AC

Sol. The nucleophiles contains two donor atoms.

22. AB

Sol. $$E_{Cell} = E^0_{Cell} - \frac{0.0591}{2}\log\frac{\left[Zn^{2+}\right]}{\left[Ag^{+}\right]^2}$$

$$= E^0_{Cell} - \frac{0.0591}{2}\log\frac{C_1}{(C_2)^2}$$

$E_{Cell} = E^0_{Cell}$, when $C_1 = C_2 = 1$ M or $C_2 = \sqrt{C_1}$

23. ABCD

Sol. $$3NH_4Cl + 3BCl_3 \xrightarrow[140-150^oC]{C_6H_5Cl} B_3N_3H_3Cl_3$$

$$B_3N_3H_6 \xleftarrow{NaBH_4} B_3N_3H_3Cl_3$$

24. AB

Sol. (D) forms acetone on heating
(C) forms anhydride on heating

25. ABD

Sol. CO_3^{2-}, SO_3^{2-}, NO_2^- form CO_2, SO_2 and NO or NO_2 gases when treated with dil. acids.

26. ABC

Sol. Heat of hydrogenation, heat of neutralisation and acid dilution are exothermic processes.

27. C

Sol. CsCl has b.c.c unit cell, which coordination number equal to eight.

28. B

Sol. In ZnS crystal, cations occupy half of tetrahedral voids and anions form the fcc lattice.

29. B

Sol. Due to half-filled $t_{2g}^3 e_g^2$ configuration $[FeF_6]^{3-}$ complex is colourless.

30. A

Sol. Due to $3d^9$ $\left(t_{2g}^6 e_g^3\right)$ configuration.

SECTION – C

31. 00418.52

Sol. Moles of $Sn^{2+} = 523.15 \times 10^{-3}$

Moles of MnO_4^- required = $\frac{2}{5} \times 523.15 \times 10^{-3}$

Volume = $\frac{\frac{2}{5} \times 523.15 \times 10^{-3} \times 1000}{0.5}$ = 418.52 mL

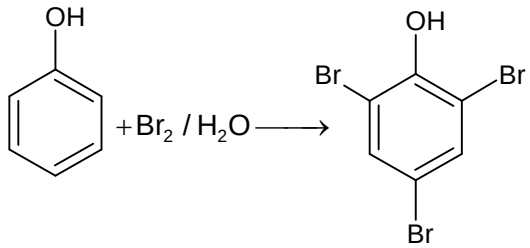
32. 00020.22

Sol. $\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9 \times 10^{-31} \times 50 \times 10^{3}}} = \frac{6.6 \times 10^{-34}}{\sqrt{900 \times 10^{-28}}}$

$= \frac{6.6 \times 10^{-34}}{30 \times 10^{-14}} = 0.22 \times 10^{-20} = y \times 10^{-z}$

$\therefore$ y + z = 20.22

33. 00330.70

Sol. 

34. 00140.80

Sol. $\Delta T_f = K_f m = 1.86 \times$ one mole $\times \frac{1000}{W}$

or $1.86 \times \frac{1000}{W} = 3.72$

or W = 500 g

$\therefore$ Mass of ice formed = 640.8 – 500 = 140.8

35. 00154.90

Sol. $X = K_2Cr_2O_7$, $Y = CrO_2Cl_2$, $Z = Na_2CrO_4$, Yellow ppt. = $PbCrO_4$

36. 02412.50

Sol. Q = It

190.5 g Cu = 3 mole

½ mole Cu deposited by 1 F

3 mole Cu deposited = 6 F = 6×96500 = 579000 coulomb

$579000 = I \times 60 \times 4$

or $I = \frac{579000}{240} = 2412.5$ amp

Mathematics

PART – III

SECTION – A

37. AD

Sol. $\because$ (x – 1) (x – 2) = 4

conjugate hyperbola (x – 1) (x – 2) = -4

$\therefore \frac{a^2}{2} = 4 \therefore a = 2\sqrt{2}$

focus of H' $\equiv \left(-2\sqrt{2}, 2\sqrt{2}\right) \& \left(2\sqrt{2}, -2\sqrt{2}\right)$

focus of H $\equiv \left(1-2\sqrt{2}, 2+2\sqrt{2}\right) \& \left(1+2\sqrt{2}, 2-2\sqrt{2}\right)$

Equation of transverse axis y – 2 = x – 1 $\Rightarrow$ x – y + 1 = 0.

38. AC

Sol. We have $\sum_{r=0}^{n} \frac{r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n-r}{{}^nC_{n-r}} = \sum_{r=0}^{\infty} \frac{n-r}{{}^nC_r} \Rightarrow \sum_{r=0}^{n} \frac{2r}{{}^nC_r} = \sum_{r=0}^{n} \frac{n}{{}^nC_r} \Rightarrow n = n^2 - 3n + 3$

$\Rightarrow n = 1, 3$

39. ABC

Sol. Combined equation of pair of lines through the origin joining the points of intersection of lien $y = \sqrt{m}x + 1$ with the given curve is $x^2 + 2xy + (2 + \sin\alpha)y^2 - (y - mx)^2 = 0$ for the chord to subtend a right angle at the origin $(1-m) + (2 + \sin^2\alpha - 1) = 0$ (as sum of the coefficient of $x^2 + y^2 = 0$)

$\Rightarrow \sin^2\alpha = m - 2$

$\Rightarrow 0 \le m - 2 \le 1$

$\Rightarrow 2 \le m \le 3$

40. ABCD

Sol. f(x) = f(α + β - x)

$\Rightarrow$ f(α) = f(β) and g(α) = g(β)

$\int_{\alpha}^{\beta} f(x)g'(x)\,dx = f(x)g(x)\Big|_{\alpha}^{\beta} - \int_{\alpha}^{\beta} f'(x)g(x)dx$

$2I = \int_{\alpha}^{\beta} f(x)g'(x) - \int_{\alpha}^{\beta} f'(x)g(x)dx$

$I = \frac{1}{2}\int_{\alpha}^{\beta} (f(x)g'(x) - f'(x)g(x))\,dx$

41. BD

Sol. Let one probability of choosing one integer k be $P(k) = \frac{\lambda}{k^4}$. ($\lambda$ is one constant of proportionality). Then,

$$\sum_{k=1}^{2m}\frac{\lambda}{k^4}=1\Rightarrow\lambda\sum_{k=1}^{2m}\frac{1}{k^4}=1$$

Let x_1 be the probability of choosing the odd number. Then

$$x_1=\sum_{k=1}^{2m}P(2k-1)=\lambda\sum_{k=1}^{m}\frac{1}{(2k-1)^4}$$

Also, $1-x_1=x_2=\sum_{k=1}^{m}P(2k)=\lambda\sum_{k=1}^{m}\frac{1}{(2k)^4}<\lambda\sum_{k=1}^{m}\frac{1}{(2k-1)^4}$

$$\Rightarrow\ 1-x_1<x_1\qquad\Rightarrow\qquad x_1>\frac{1}{2}\qquad\Rightarrow\qquad x_2<\frac{1}{2}$$

42. BCD

Sol. $f''(x)\Rightarrow\begin{cases}2\,;\,x>0\\-2\,;\ x<0\end{cases}\Rightarrow$ not differentiable

$$g(x)\Rightarrow\begin{cases}-x^2\cot^{-1}x;\quad x>0\\-x^2\cot^{-1}x;\quad x<0\\\qquad 0\ ;x=0\end{cases}\Rightarrow\text{ twice differentiable}$$

$h(x)=\sin^2x\ \rightarrow$ differentiable

$$k'(x)=\begin{cases}4x^3.\cos\frac{1}{x}+x^2.\sin\frac{1}{x}\,;\,x\neq0\\\qquad\qquad 0\,;\ x=0\end{cases}$$

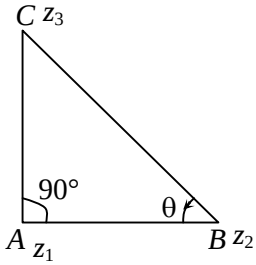
twice differentiable

43. AB

Sol. $\dfrac{W_2}{|W_2|}=\dfrac{W_1}{|W_1|}e^{-i\theta}$

$$\Rightarrow\qquad\frac{W_2}{W_1}=\frac{|W_2|}{|W_1|}e^{-i\theta}=1-i\tan\theta$$

$$W_1\overline{W}_2=\frac{W_1}{W_2}|W_2|^2=|W_1|\ |W_2|e^{i\theta}=\left(|W_1|^2+i|W_1|\ |W_2|\tan\theta\right)$$



44. ABC

Sol. We have $A(\alpha,\beta)'=\begin{bmatrix}\cos\alpha&-\sin\alpha&0\\\sin\alpha&\cos\alpha&0\\0&0&e^{\beta}\end{bmatrix}$

$$=\begin{bmatrix}\cos(-\alpha)&\sin(-\alpha)&0\\-\sin(-\alpha)&\cos(-\alpha)&0\\0&0&e^{\beta}\end{bmatrix}=A(-\alpha,\beta)$$

Also, $A(\alpha,\beta)A(-\alpha,-\beta)$

$$= \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & e^{\beta} \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & e^{-\beta} \end{bmatrix}$$

$= I$

$\Rightarrow A(\alpha,\beta)^{-1} = A(-\alpha, -\beta)$

Next, $\text{Adj}(\alpha,\beta) = |A(\alpha,\beta)| A(\alpha, \beta)^{-1}$

$= e^{\beta} A(-\alpha, -\beta)$

45. C
46. C

Sol. (for Q. 45 – 46):

(I). Equation of the plane π is $A(x-1)+B(y-1)+C(z-1)=0$ where $A+0$. B –1, C = 0 and $-A+B+0.C=0$

$\Rightarrow A=B=C$ and the equation of π is $x+y+z=3x$. Which meets the coordinates axes at A (3, 0, 0), B (0, 3, 0) and C (0, 0, 3). So volume of the tetrahedron

$$\text{OABC} = \frac{1}{6}\begin{vmatrix} 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & 1 \\ 0 & 3 & 0 & 1 \\ 0 & 0 & 3 & 1 \end{vmatrix} = \frac{1}{6} \times 27 = \frac{9}{2}$$

(II). $AB = BC = CA = 3\sqrt{2}$, so the area of the face ABC which is an equilateral triangle $= \frac{\sqrt{3}}{4} \times \left(3\sqrt{2}\right)^2 = \frac{9\sqrt{3}}{2}$. Each of the face OAB, OBC, OCA is an isosceles triangle with sides $3, 3\sqrt{2}$

So area of each of these faces is $\frac{1}{2} \times 3\sqrt{2} \times \sqrt{9 - \frac{9}{2}} = \frac{9}{2}$

(III). required distance is $\left|\frac{3\sqrt{3}+3\sqrt{3}-3-3}{\sqrt{1+1+1}}\right| = \frac{9}{2}$

(IV). Distance of O from the face $\text{ABC} = \left|\frac{0+0+0-3}{\sqrt{1+1+1}}\right| = \sqrt{3}$ distance of A, B, C from opposite faces OBC, OCA, OAB are equal and is 3.

47. B
48. B

Sol. (for Q. 47 – 48):

a, b, c are three roots of the quadratic equation

$\left(\left(4f(-1)-3\right)x^2 + \left(4f(1)-3\right)\right)x + f(2) = 0$

$\Rightarrow 4f(-1) = 3, 4f(1) = 3$ and $f(2) = 0$

Let $f(x) = (x-2)(Ax+B)$

Now, $3 = 4f(-1) = 4(-3)(-A+B) \Rightarrow A - B = \frac{1}{4}$

$3 = 4f(-1) = 4(-1)(A+B) \Rightarrow A+B = \dfrac{-3}{4}$

$\therefore\ A = \dfrac{-1}{4}, B = \dfrac{-2}{4}$

Thus, $f(x) = \dfrac{1}{4}(4-x^2)$.

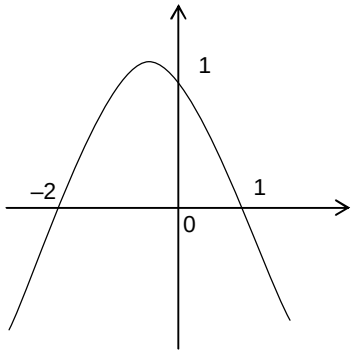
The graph of $y = f(x)$ is given in Figure.

(I). x coordinates of points of intersection of $y = f(x)$ when the x – axis are ± 2.

(II). Area $= \dfrac{3}{2}\int_{-2}^{2} \dfrac{1}{4}(4-x^2)dx = \dfrac{3}{8}(2)\left(4x - \dfrac{x^2}{3}\right)\Big]_0^2 = 4$

(III). Maximum value of f(x) is 1

(IV). Length of intercept on the x – axis is 4.



SECTION – C

49. 00021.61

Sol. The number of ways when no student failed in any examination $= (2^3-1)^4$

The number of ways when out of above cases atleast one subject was not cleared by any students $= 3C_1(3)^4$

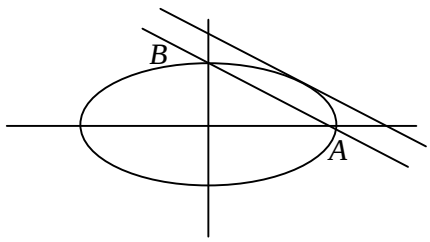
The number of ways when out of above cases atleast any two subjects were not cleared by any student $= 3C_2(1)^4$. So required cases $= (2^3-1)^4 - 3C_1 3^4 + 3C_2$ = 2161

50. 00003.00

Sol. Let perpendicular distance of *P* from the line be *h*

$\dfrac{1}{2}\times h \times 5 = 6(\sqrt{2}-1)$ (as $\Delta PAB = 6(\sqrt{2}-1)$)

$\Rightarrow h = \dfrac{12(\sqrt{2}-1)}{5}$



Now distance of tangent parallel to *AB* i.e. $4y + 3x = 12\sqrt{2}$, from line *AB* is $\dfrac{12(\sqrt{2}-1)}{5}$.

There are just three such points.

51. 00013.33

Sol. Let $\vec{r}=\lambda\vec{b}+\mu\vec{c}$ and $\vec{c}=\pm(xi+yj)$. Since $\vec{c}$ and $\vec{b}$ are perpendicular, we have

$4x+3y=0 \quad \Rightarrow \quad \vec{c}=\pm x\left(i-\frac{4}{3}j\right)$

$\pm 1=$ proj. of $\vec{r}$ on $\vec{b}=\frac{\vec{r}.\vec{b}}{|\vec{b}|}=\frac{(\lambda\vec{b}+\mu\vec{c}).\vec{b}}{|\vec{b}|}=\frac{\lambda\vec{b}.\vec{b}}{|\vec{b}|}$

$\left[\because \vec{b}.\vec{c}=0\right]$

$=\lambda|\vec{b}|=5\lambda$. Hence $\lambda=\pm\frac{1}{5}$

Also, $\pm 2=$ proj. of $\vec{r}$ on $\vec{c}=\frac{\vec{r}.\vec{c}}{|\vec{c}|}$

$=\frac{(\lambda\vec{b}+\mu\vec{c}).\vec{c}}{|\vec{c}|}=\mu|\vec{c}|=\frac{5}{3}\mu x$

Thus, $\mu x=\pm\frac{6}{5}$. Therefore,

$\vec{r}=\frac{1}{5}(4i+3j)+\frac{6}{5}\left(i-\frac{4}{3}j\right)=\pm(2i-j)$

$\vec{r}=\frac{1}{5}(4i+3j)-\frac{6}{5}\left(i-\frac{4}{3}j\right)=\pm\left(-\frac{2}{3}i+\frac{11}{5}j\right)$

Thus there are four such vectors

$\sum_{i=1}^{4}|\vec{r_i}|^2=2|2i-j|^2+2\left|-\frac{2}{5}i+\frac{11}{5}j\right|^2=20$

52. 00000.50

Sol. Put $\frac{1}{x}=t$, so that $I=\int_{1/e}^{e}|\log x|(-1)\,dt$

$=\int_{e}^{1/e}\left|\log\frac{1}{t}\right|(-1)\,dt$

$=\int_{1/e}^{e}|-\log t|\,dt$

$=\int_{1/e}^{1}(-\log t)\,dt+\int_{1}^{e}(\log t)\,dt$

$=(-t\log t+t]_{1/e}^{1}+(t+\log t-t]_{1}^{e}$

$=1-\frac{1}{e}-\frac{1}{e}+e-e+1=2\left(1-\frac{1}{e}\right)$

53. 00005.24

Sol. Given, $f(x)+f(y)=f\left(\frac{x+y}{1-xy}\right)$

Putting x = 0, y = 0, we get f (0) = 0 ……..(i)

and putting y = –x, we get

$f(x)+f(-x)=f(0)=0$

$\therefore \quad f(x)=-f(-x)$ ……(ii)

Now, $f'(x)=\lim_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

$=\lim_{h\to 0}\dfrac{f(x+h)+f(-x)}{h}$ [from Eq. (ii)]

$=\lim_{h\to 0}\dfrac{f\left(\dfrac{x+h-x}{1+(x+h)x}\right)}{h}$

$=\lim_{h\to 0}\dfrac{f\left(\dfrac{h}{1+x^2+xh}\right)}{\left(\dfrac{h}{1+x^2+xh}\right).(1+x^2+xh)}$

$\Rightarrow \quad f'(x)=\dfrac{2}{1+x^2}$ $\left(\because \lim_{x\to 0}\dfrac{f(x)}{x}=2\right)$

$\therefore f'(-2)=\dfrac{2}{1+4}=\dfrac{2}{5}$

and $f(x)=2\tan^{-1}x+c$

$f(0)=0+c=0$

$\therefore \quad c=0$

$\therefore \quad f(x)=2\tan^{-1}x$

$\Rightarrow \quad f\left(\sqrt{3}\right)=2\tan^{-1}\sqrt{3}=\dfrac{2\pi}{3}$

54. 00007.00

Sol. The given expression can be interpreted as the square of the distance between the points $(\tan A, \cot A)$ and $(\cos B, \sin B)$.

The minimum value of this distance is the minimum distance between the curves $xy=1$ and $x^2+y^2=1$.